

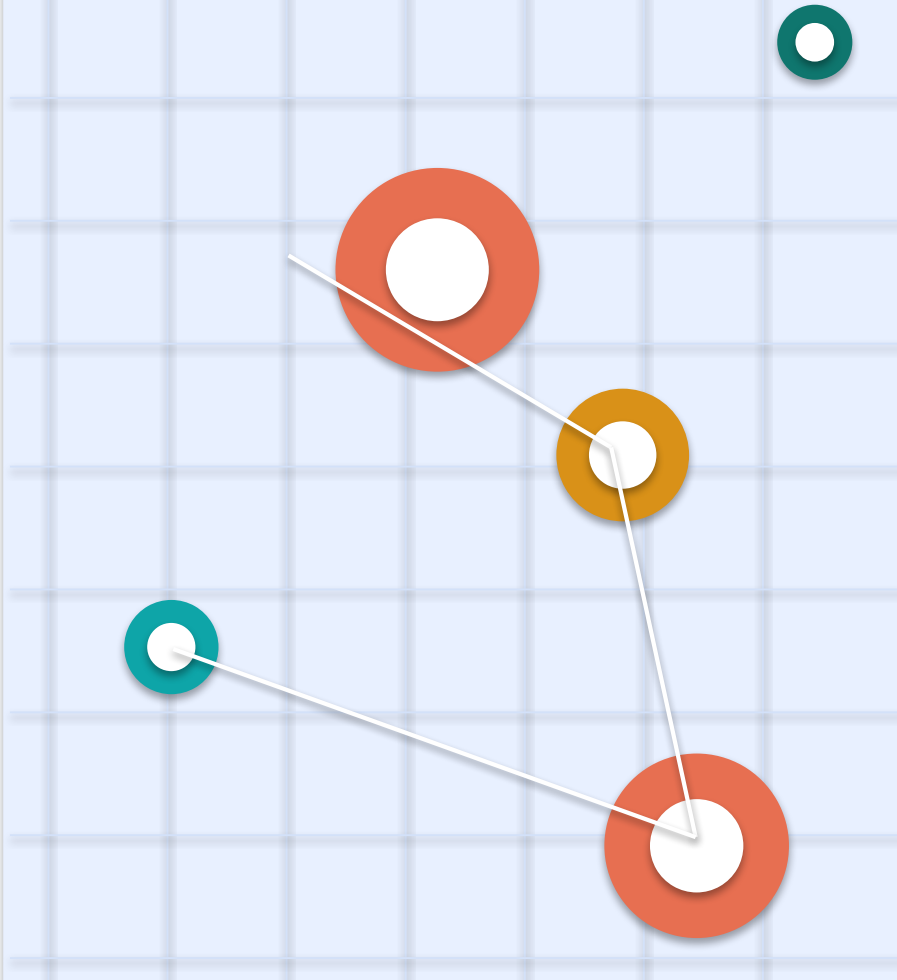
Predictive Cybercrime Intelligence

Forecasting where illicit cash may surface — before the next withdrawal happens.

A spatio-temporal AI framework that turns scattered complaints into an early-warning signal for investigators.

SIH26184 · MINISTRY OF HOME AFFAIRS

TEAM NAME: TRACE X



From scattered reports to a risk map



OUR SOLUTION

Turn complaints into a live risk map

Complaints scatter after cash-out begins — this system scores each cell, combines three signals, and flags where cash may surface next.

ONE UNIFIED SIGNAL

Where is cash most likely to surface?

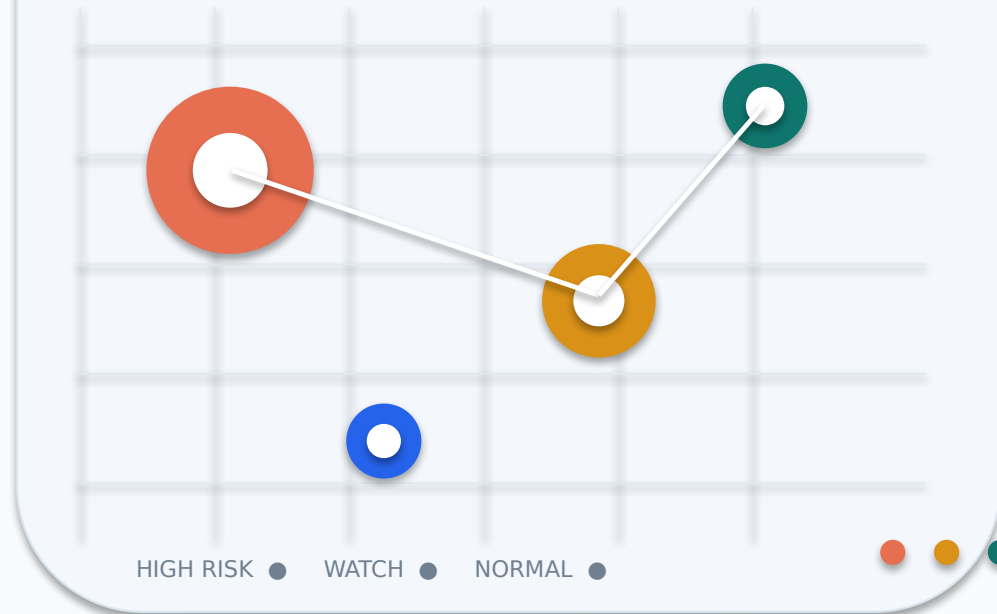
A live, self-updating view built for investigators — not just analysts.

HOTSPOTS

PATTERNS

ANOMALIES

PREDICTIVE RISK MAP · NEXT 6 HOURS



WHAT INVESTIGATORS GET

Prioritized cells

Ranked locations and time windows.

Reason codes

The pattern behind every alert.

Filterable view

Focus by area, time or risk level.



Three models, one practical pipeline to action

Hotspot scoring, pattern clustering and anomaly detection combine in an upgrade-ready pipeline.

END-TO-END INTELLIGENCE PIPELINE



Prototype today

Grid-cell binning and Flask stand in for PostGIS and FastAPI; the API contract remains ready for a drop-in upgrade.



A safe prototype pathway to operational validation

The workflow is demonstrated on synthetic records; real performance metrics should be measured only after authorized agency data is available.

Prototype

Safe test scope

Demonstrated on privacy-safe synthetic records.

Signals

Pattern validation

The workflow surfaces repeatable patterns for review.

Next step

Agency evaluation

Measure lead time and alert quality with authorized data.

DATA STRATEGY



Synthetic records

Prove the pipeline safely



Open data

Test broader signals



Authorized agency data

Operationalize with controls

Prototype finding: unsupervised clustering recovered simulated fraud hotspots without labeled training data.



Faster signals, fewer losses, real scale

Earlier detection and prioritized leads — with privacy and human oversight built in.

KEY BENEFITS

01 Faster detection

Cuts the gap between complaint and action from days to hours.

02 Reduced losses

Earlier alerts mean more illicit cash is intercepted before it disappears.

03 Built to scale

Thousands of complaints resolve into a handful of prioritized leads.

SAFEGUARDS

Decision-support, not verdict.

Privacy-by-design. Human oversight.
No automated punitive action.

PRIVACY & OVERSIGHT BY DESIGN



Built on established open-source tools and methods

The prototype combines proven open-source libraries with published ML techniques, tested only on privacy-safe synthetic data.

Libraries

Open-source stack

GeoPandas, scikit-learn, FastAPI, React.

Methods

ML techniques used

Gradient boosting, DBSCAN clustering, Isolation Forest.

Data

Synthetic-only note

No real complaint or transaction records were used.

KEY REFERENCES

01

GeoPandas & scikit-learn

Core spatial and ML libraries

02

DBSCAN & Isolation Forest

Clustering and anomaly detection

03

FastAPI & React

Serving API and investigator UI

Note: this prototype uses only privacy-safe synthetic data — no real complainant or suspect information.